EXERCISES ABOUT CHAPTER 13 Properties of Solutions

example : 10 mL C_2H_5OH ($d = 0.789$ g/mL) dissolved in water and diluted to 100 mL volume. An aq solution was prepared with $d = 0.982$ g/mL

Calculate a) % volume of C_2H_5OH
b) mass %
c) mole fraction
d) molarity M
e) molality

$$ppm = \frac{1\text{ g solute}}{1\,000\,000\text{ g solv}}$$

$$ppb = \frac{1\text{ g solute}}{1\,000\,000\,000\text{ g solv}}$$

a) $$\text{volume \%} = \frac{10\text{ mL ethanol}}{100\text{ mL solution}} \times 100\,\% = 10\,\%$$

b) $$\text{mass \%} = \frac{10\text{ mL} \times \frac{0.789\text{ g/mL}}{1\text{ mL}}}{100\text{ mL} \times \frac{0.982\text{ g/mL}}{1\text{ mL}}} 100\,\% = \frac{7.89\text{ g ethanol}}{98.2\text{ g solution}} \times 100\,\%$$

$$= 8.03\,\%$$

c)

$$\frac{7.89\ g\ \text{etanol}}{46.07\ g/mol} = 0.171\ mol\ \text{etanol}$$

$98.2\ g - 7.89\ g = 90.3\ g$ su

$$\frac{90.3\ g\ su}{18.02\ g/mol} = 5.01\ mol\ su$$

$$X_{C_2H_5OH} = \frac{0.171}{0.171 + 5.01} = 0.033$$

d) $$M = \frac{0.171\ mol\ C_2H_5OH}{0.1\ L\ solution} = 1.71\ M\ C_2H_5OH$$

(2)

e) $$m = \frac{\text{moles of solute}}{kg\ of\ solvent} = \frac{0.171\ mol\ C_2H_5OH}{0.0903\ kg\ H_2O} = 0.189\ m\ C_2H_5OH$$

$$90.3\ g\ H_2O \times \frac{1\ kg\ H_2O}{1000\ g\ H_2O} = 0.0903\ kg$$

* Henry's Law : $S_g = k \cdot P_g$

example : At 0°C and 1 atm O_2 pressure the solubility of $O_2(g)$ is 48.9 mL/L.

At normal partial pressure of oxygen in air (0.2095 atm) what is the concentration of the saturated aqueous solution?

0°C and $P_{O_2} = 1$ atm

$$M = \frac{0.0489 \text{ L } O_2 \times \frac{1 \text{ mol } O_2}{22.4 \text{ L } O_2 \text{ (STP)}}}{1 \text{ L solution}} = 2.18 \times 10^{-3} \text{ M } O_2$$

According to Henry's Law $S_g = k \cdot P_g$

$$S_g = \frac{2.18 \times 10^{-3} \text{ M } O_2}{1 \text{ atm}} \times 0.2095 \text{ atm} = 4.57 \times 10^{-4} \text{ M } O_2$$

to reach k constant we should know the solubility of a specific gas at a known pressure and temperature.

* Colligative properties : vapor-pressure lowering
boiling-point elevation
melting-point depression
osmotic pressure

* Raoult's Law : $P_A = X_A \cdot P_A^\circ$

example : The vapor pressures of pure bentene and toluen at 25°C are 95.1 mmHg and 28.4 mmHg, respectively. A solution was prepared with both mole fractions are 0.5

$X_{bentene} = 0.5$ $X_{toluen} = 0.5$

Calculate the total vapor pressure of solution.

$P_{bentene} = X_{bentene} \cdot P^\circ_{bentene} = 0.5 \times 95.1 \text{ mmHg} = 47.6 \text{ mmHg}$

$P_{toluen} = X_{toluen} \cdot P^\circ_{toluene} = 0.5 \times 28.4 \text{ mmHg} = 14.2 \text{ mmHg}$

Total vapor pressure = 61.8 mmHg

* Boiling-point elevation $\Delta T_b = K_b \cdot m$

Freezing-point depression $\Delta T_f = -K_f \cdot m$

example : Nicotine is soluble in water under 60°C.

a) What is the molality of an aq. solution of nicotine with a freezing point −0.45°C ?

b) If this solution is prepared by dissolving 1.921 g nicotine in 48.92 g water, what is the molecular weight of nicotine?

c) Burning analysis result for nicotine gave 74.03 % C
8.70 % H
17.27 % N

what is the molecule formula of nicotine ?

(4)

a) $K_f = 1.86\ ^\circ C/m \rightarrow$ molal

$$\Delta T_f = -K_f \cdot m \quad \Rightarrow \quad m = \frac{\Delta T_f}{K_f} = \frac{-0.450\ ^\circ C}{-1.86\ ^\circ C/m} = 0.242 \text{ molal}$$

$0.242\ m$

$\Delta T_f = -0.450\ ^\circ C - 0\ ^\circ C = -0.450\ ^\circ C$

b) molalite $\rightarrow$ $\dfrac{\text{mol of solute}}{\text{kg of solvent}}$

$$\frac{0.242 \text{ mol nicotine}}{\text{kg water}} = \frac{1.821\ g/mw}{0.0482 \text{ kg water}}$$

$mw \cong 162\ g/mol$

c) 100 g nicotine sample

$$\frac{74.03}{12.011} = 6.16 \qquad \frac{8.70}{1.008} = 8.63 \qquad \frac{17.27}{14.007} = 1.23$$

$$C_{\frac{6.16}{1.23}} H_{\frac{8.63}{1.23}} N_{\frac{1.23}{1.23}} \longrightarrow C_5 H_7 N$$

calculate molecular welpht $\rightarrow$ $(5 \times 12.011) = 60.055$

$(7 \times 1.008) = 7.056$

$+ \quad 14.007$

$\cong 81$

(6) $\rightarrow$ 162 g/mol mw

$\Rightarrow$ $C_{10} H_{14} N_2$

* Electrolytes : Van't Hoff factor

Suppose there is 0.01 molal aq solution

freezing-point decrease would be $\Delta T_f = -K_f \cdot m$

$\Delta T_f = -1.86\ ^\circ C/m \cdot 0.01\ m$

$\Delta T_f = -0.0186^\circ C$

If this is urea solution the freezing point is measured as $-0.0186^\circ C$

but if this is 0.01 m NaCl solution the freezing point $-0.0361^\circ C$

Van't Hoff factor $i = \dfrac{-0.0361^\circ C}{-1.86\ ^\circ C/m \times 0.01\ m}$ $\dfrac{\text{measured } \Delta T_f}{\text{expected } \Delta T_f} = 1.94$

example

urea, glycerine, glycose → non electrolytes $i = 1$

NaCl → electrolyte $i = 2$ 1 mol compound ≤ 1 Na^+, 1 Cl^- two ions

$MgCl_2$ → $i = 3$

* Osmosis - Osmotic pressure

Çözelti üzerine yeteri kadar büyük basınç uygulanırsa suyun net geçişi sıfırlanabilir Osmotik geçişi durdurmak için çözeltiye uygulanması gereken basınca çözeltinin osmotik basıncı denir.

$$\pi = \left(\frac{n}{V}\right) \cdot RT$$

$$\pi = M \cdot RT$$

example: Calculate the osmotic pressure of 0.001 M sucrose solution at 25°C

$$\pi = 0.001\ M \cdot 0.082 \frac{L \cdot atm}{mol \cdot K} \cdot 298\ K$$

$$\pi = 0.024\ atm$$